

Technology Stack

Date	04 July 2026
Team ID	SWTID-2026-7391
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, Bootstrap	Provides a simple, responsive, and user-friendly interface for entering credit card application details and displaying prediction results.

2	Backend / Server-Side	Python, Flask	Processes user requests, loads the Machine Learning model, and generates credit card approval predictions efficiently.
3	Database / Data Storage	CSV Dataset / SQLite	Stores customer application data and is used for training and testing the Machine Learning model.
4	Cloud / Hosting / Deployment	Streamlit Cloud / Render	Hosts the application online, allowing users to access the prediction system from anywhere.
5	Version Control & CI/CD	GitHub	Manages source code, tracks project changes, and enables team collaboration.

6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Joblib	Scikit-learn builds the ML model, Pandas handles data processing, NumPy performs numerical computations, and Joblib saves and loads the trained model